

BattleReef BRMC Consumer v1.0

Prototype electrical engineering review package

ENGINEERING REVIEW RELEASE - NOT FABRICATION AUTHORITY

Document BRMC-ELEC-EVT-030 | Revision C | 2026-09-03

Review object: 220 x 78 mm, six-layer modular passive backplane; official CM5 IO Board rev2 prototype endpoint; controlled harness and enclosure-base interface evidence.

Release statement

This package is complete for independent engineering review of the stated backplane scope. It is not a fabrication authorization and does not approve absent daughtercard or PSM-01 schematics. Fabricator DFM, first-article checks, measured EVT evidence, and signed qualified review remain mandatory gates.

1. Configuration and evidence

The deterministic PCB generator, generated KiCad board, rules, pinout, connector schedule, interconnect drawing, power budget, mechanical base candidate, and fail-closed validator are the controlled source. Accept manufacturing outputs only from a successful KiCad 9 CI run for the exact reviewed revision. The production-connector layout baseline recorded in release-gate-evidence.json passed KiCad 9.0.9 with zero DRC violations, zero unconnected pads and zero footprint errors; verify its commit, run, artifact ID and SHA-256 before review disposition.

Evidence	Purpose
BRMC_Consumer_EVT_Backplane_v1.0.kicad_pcb	reviewed PCB source
BRMC_Consumer_EVT_Backplane_v1.0.kicad_dru	controlled rules
BRMC_Consumer_EVT_Backplane_v1.0_Interconnect_Schematic.pdf	schematic-equivalent connectivity
BRMC_Consumer_EVT_Backplane_v1.0_Netlist.csv	machine-readable net/endpoints
BRMC_Consumer_v1.0_Power_Budget.csv	load ceilings and margins
BRMC_Consumer_v1.0_Connector_and_Harness_Schedule.csv	wire/contact/drop/protection schedule
BRMC_Consumer_v1.0_Enclosure_Base_6Boss_CNC6061_RELEASE_CANDIDATE.step	base CAD candidate
..._Drawing.pdf / dimensions.csv / interference.csv	boss dimensions and nominal CAD evidence
independent-review-checklist.csv	review return record

2. Architecture decisions

EVT uses the official Raspberry Pi CM5 IO Board revision 2 outside the enclosure, independently powered at J11 by an official 27 W USB-C supply. J_CM5 carries GPIO and four returns only. A custom integrated carrier is deferred to a new revision. The current KiCad board is a passive modular backplane; it does not contain the integrated v0.9 power, safety, bus, analog or rear-I/O circuitry. Those circuits remain separate module approval objects.

Boundary	Controlled decision	Reviewer focus
CM5	external CM5IO rev2; independent 5 V / 5 A envelope	GPIO voltage, boot strapping, no rail backfeed
Power	24 V / 2.5 A source; PSM-01 converts/protects; interface ceilings below	fault coordination and missing PSM schematic
Backplane	passive connectors/tracks/planes only	pin map, copper/vias, returns, connector footprints
Enclosure	CNC 6061 base Rev B candidate with six M3 bosses	clearance/tolerance evidence and approval boundary

3. Worst-case power budget

Values labeled design allowance are mandatory interface ceilings. They are conservative allocations, not fabricated measurements. The 24 V source calculation uses 88% conversion efficiency for 5 V and 12 V branches and a 25% startup/transient multiplier.

Rail	Endpoint	V	Expected A	Worst A	Allowance A	Protection/status
CM5IO 5V	CM5 + CM5IO local loads	5.0	0.9	5.0	5.0	Official 27 W USB-C PSU and CM5IO input protection
DISPLAY_5V	Waveshare 10.1inch DSI LCD (C)	5.0	0.52	0.8	1.0	PSM-01 dedicated 1.25 A protected branch
5V_SYS	J_MCU safety-MCU module	5.0	0.1	0.2	0.25	PSM-01 5V_SYS 1.25 A branch limit
5V_SYS	J_PH isolated pH carrier	5.0	0.015	0.05	0.1	Module OFF control; shared 5V_SYS branch
5V_SYS	J_ORP isolated ORP carrier	5.0	0.015	0.05	0.1	Module OFF control; shared 5V_SYS branch
5V_SYS	J_EC isolated EC carrier	5.0	0.015	0.05	0.1	Module OFF control; shared 5V_SYS branch
5V_SYS	J_CAN isolated CAN FD module	5.0	0.08	0.15	0.2	Daughtercard input fuse/current limit required
5V_SYS	J_485 isolated RS-485 module	5.0	0.08	0.15	0.2	Daughtercard input fuse/current limit required
3V3_SYS	J_TEMP + J_SVC reference	3.3	0.03	0.08	0.1	100 mA current-limited output; no external backfeed
12V_SYS	reserved prototype auxiliary	12.0	0.0	0.2	0.25	PSM-01 0.50 A protected branch
24V_IN	J_AO 8-channel 0-10 V module electronics	24.0	0.12	0.2	0.25	0.35 A daughtercard branch protection required
24V_IN	J_PWRMOD external power-module electronics	24.0	0.25	0.4	0.5	0.75 A branch fuse/eFuse required
24V_IN	J_SAFE relay-driver electronics	24.0	0.04	0.08	0.1	0.20 A branch protection and flyback on daughtercard

Calculated 24 V envelope: direct 24 V loads 0.85 A + (5 V backplane 1.00 A / 88%)×5/24 + (display 1.00 A / 88%)×5/24 + (12 V 0.25 A / 88%)×12/24 = 1.47 A continuous. With 25% margin: 1.83 A. Adapter headroom: 0.67 A to the 2.5 A ceiling.

4. Harness schedule

ID	Source → destination	Function	V / allowance	Conductor / max length	Drop	Protection
HARN-01	Phoenix Contact M12 panel inlet 1419742 → PSM-01 protected power module	24 V system input	24 V / 2.5 A	20 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.5 m	0.101 V (0.42%)	PSM input eFuse target 2.35 A; Mean Well source 2.5 A
HARN-02A	PSM-01 → Backplane J_PWR	protected 24 V rail	24 V / 0.85 A	20 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.35 m	0.024 V (0.10%)	1.0 A protected branch
HARN-02B	PSM-01 → Backplane J_PWR	5V_SYS	5 V / 1.0 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.35 m	0.045 V (0.90%)	1.25 A protected branch
HARN-02C	PSM-01 → Backplane J_PWR	reserved 12V_SYS	12 V / 0.25 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.35 m	0.011 V (0.09%)	0.50 A protected branch
HARN-03	CM5IO J17 → Waveshare display DSI	MIPI DSI signal	0 V / 0 A	signal cable; see construction; 0.2 m	N/A V (N/A%)	not applicable
HARN-04	PSM-01 dedicated 5 V output → Waveshare display power input	display power	5 V / 1.0 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.5 m	0.064 V (1.29%)	dedicated 1.25 A current-limited branch
HARN-CM5-SIG	Backplane J_CM5 → CM5IO J8	3.3 V GPIO and four returns; no power	3.3 V / 0 A	signal cable; see construction; 0.3 m	N/A V (N/A%)	no 5 V or 3V3 contacts fitted at J8
HARN-MCU-PWR	Backplane J_MCU → Safety MCU daughtercard	MCU module power/3V3 ownership	5 V / 0.25 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.010 V (0.19%)	shared 5V_SYS; daughtercard 3V3 output limited 0.10 A
HARN-PH	Backplane J_PH → Atlas isolated pH carrier	pH carrier power/data	5 V / 0.1 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.004 V (0.08%)	shared 5V_SYS; OFF provides controlled disable
HARN-ORP	Backplane J_ORP → Atlas isolated ORP carrier	ORP carrier power/data	5 V / 0.1 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.004 V (0.08%)	shared 5V_SYS; OFF provides controlled disable
HARN-EC	Backplane J_EC → Atlas isolated conductivity carrier	conductivity carrier power/data	5 V / 0.1 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.004 V (0.08%)	shared 5V_SYS; OFF provides controlled disable
HARN-TEMP	Backplane J_TEMP → digital temperature interface	digital temperature	3.3 V / 0.05 A	24 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.5 m	0.005 V (0.16%)	3V3 source current-limited on MCU module
HARN-CAN	Backplane J_CAN → isolated CAN daughtercard	CAN module power and bus	5 V / 0.2 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.008 V (0.15%)	daughtercard input branch protection
HARN-485	Backplane J_485 → isolated RS-485 daughtercard	RS-485 module power and bus	5 V / 0.2 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.008 V (0.15%)	daughtercard input branch protection
HARN-AO	Backplane J_AO → 0-10 V daughtercard	module power and Modbus	24 V / 0.25 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.35 m	0.011 V (0.05%)	0.35 A daughtercard branch protection
HARN-PWRMOD	Backplane J_PWRMOD → external power-module electronics	module power/bus/safety	24 V / 0.5 A	20 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.35 m	0.014 V (0.06%)	0.75 A branch fuse/eFuse
HARN-SAFE	Backplane J_SAFE → safety relay driver	safe-state relay driver	24 V / 0.1 A	22 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.3 m	0.004 V (0.02%)	0.20 A protected branch; flyback at driver
HARN-SVC	Backplane J_SVC → service fixture	fixture-only debug	3.3 V / 0.05 A	26 AWG stranded Cu, UL1007 or equivalent, 105 C; 0.2 m	0.003 V (0.10%)	3V3 limited to 100 mA at source; no target backfeed
HARN-05	CM5 RF connector → enclosure antenna position	Wi-Fi/Bluetooth	0 V / 0 A	signal cable; see construction; 0.2 m	N/A V (N/A%)	not applicable

5. Production connector implementation

Ref	Board MPN	Circuit / pitch	Orientation	Mate/contact	Rating	Statu
J_CM5	Molex 90130-1216	2x8 / 16; 2.54 mm	vertical top-entry	90142-0016 + 90119-2121; CM5IO end 90142-0040 + 90119-2121	350 V; 3 A/contact; -55..125 C header	DES
J_MCU	Molex 90130-1224	2x12 / 24; 2.54 mm	vertical top-entry	90142-0024; 90119-2110 power / 90119-2121 signal	350 V; 3 A/contact; -55..125 C header	DES
J_PH	Molex 22-23-2051	1x5; 2.54 mm	vertical top-entry	22-01-3057 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_ORP	Molex 22-23-2051	1x5; 2.54 mm	vertical top-entry	22-01-3057 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_EC	Molex 22-23-2051	1x5; 2.54 mm	vertical top-entry	22-01-3057 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_TEMP	Molex 22-23-2041	1x4; 2.54 mm	vertical top-entry	22-01-3047 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_PWR	Molex 43045-0600	2x3 / 6; 3.00 mm	right-angle toward lower edge	43025-0600 + 43030-0007	600 V; header 8.5 A/contact; terminal/wire limits apply	DES
J_CAN	Molex 22-23-2061	1x6; 2.54 mm	vertical top-entry	22-01-3067 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_485	Molex 22-23-2071	1x7; 2.54 mm	vertical top-entry	22-01-3077 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_AO	Molex 43045-0400	2x2 / 4; 3.00 mm	right-angle toward lower edge	43025-0400 + 43030-0007	600 V; header 8.5 A/contact; terminal/wire limits apply	DES
J_PWRMOD	Molex 43045-0800	2x4 / 8; 3.00 mm	right-angle toward lower edge	43025-0800 + 43030-0007	600 V; header 8.5 A/contact; terminal/wire limits apply	DES
J_SAFE	Molex 22-23-2041	1x4; 2.54 mm	vertical top-entry	22-01-3047 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES
J_SVC	Molex 22-23-2081	1x8; 2.54 mm	vertical top-entry	22-01-3087 + 08-50-0032	500 V; 4 A/contact; -40..80 C	DES

Footprint ECO: J_PWR/J_AO/J_PWRMOD now use the correct 3.00 mm dual-row Micro-Fit pattern with manufacturer locator holes. Signal connectors use controlled C-Grid III or KK 254 geometry. The independent reviewer must compare pad/drill/body/courtyard/pin-1 details with the listed manufacturer drawings and verify first-article mating orientation.

6. Layout and signal-integrity review points

Topic	Controlled design / required review
Ground	L2 and L5 continuous GND zones; every GND PTH connects directly; verify antipads and return continuity in filled Gerbers.
Fan-out	Top/bottom connector classes use separated copper layers; through-via slots are clearance checked by the deterministic router.
Power copper	24 V B.Cu trunk 1.50 mm; 5 V B.Cu trunk 2.00 mm with 0.80 mm fan-outs and 0.40 mm drills; review against allowed currents and fabricator copper.
Field buses	This board is a low-speed interconnect, not a controlled-impedance CAN/RS-485 trunk. Enforce short internal stubs and twisted external pairs.
I2C	Multiple module stubs require one pull-up owner and measured bus capacitance/rise time at assembled length.
Safety	SAFETY_ENABLE is an interface signal only; independent hardware de-energize-to-safe behavior belongs on reviewed MCU/safety daughtercard.
ERC	Not applicable to the board-only passive artifact; connectivity is controlled by the generated interconnect drawing/netlist and IPC-D-356 reconciliation.

7. Mechanical evidence and limits

The enclosure base is CNC-machined 6061-T6 with six integral OD 10.00 ±0.10 mm bosses, support plane Z=8.00 ±0.05 mm, M3x0.5-6H blind threads and the controlled H1-H6 axes. The hash-controlled v0.8 assembly nominal check reports zero positive-volume intersections across 114 boss-to-nonbase-solid comparisons. The supplied assembly lacks populated daughtercard, harness, thermal and service-tool solids; therefore the independent mechanical approval and tolerance-aware first-article fit check remain required.

Hole	PCB-centered X,Y mm	Enclosure X,Y mm	PCB hole	Boss/thread
H1	-103,-32	-103,25.5	3.20 NPTH	OD10; M3x0.5
H2	0,-32	0,25.5	3.20 NPTH	OD10; M3x0.5
H3	+103,-32	+103,25.5	3.20 NPTH	OD10; M3x0.5
H4	-103,+32	-103,89.5	3.20 NPTH	OD10; M3x0.5
H5	0,+32	0,89.5	3.20 NPTH	OD10; M3x0.5
H6	+103,+32	+103,89.5	3.20 NPTH	OD10; M3x0.5

8. Review action register

ID	Owner	Required evidence	Blocks
EE-01	Qualified electrical/layout engineer	signed checklist; approved or approved-with-closed-actions	backplane fabrication release
EE-02	Electrical engineer	PSM-01 and every daughtercard schematic/BOM/layout reviewed as separate objects	complete product release
ME-01	Mechanical engineer	signed base drawing/CAD and tolerance-aware populated fit review	Item 1 closure
FAB-01	PCB fabricator	returned stackup and DFM report tied to exact commit/artifact	fabrication release
FA-01	Build owner + reviewer	connector keying/pin-1/mating access and unshrouded CM5IO J8 first-article record	Item 2/connector closure
EVT-01	Test engineer	startup, steady, peak and fault-current captures; rail droop; fuse/eFuse trip evidence	Item 2/system validation
EVT-02	Test engineer	thermal, EMC/ESD and ingress evidence under controlled configurations	product qualification

9. Independent review disposition

Reviewer name / organization: _____

Qualifications and relevant experience: _____

Reviewed git commit: _____ CI run/artifact: _____

Disposition (select one): APPROVED FOR BACKPLANE EVT FABRICATION / APPROVED WITH CLOSED ACTIONS / REJECTED

Open findings or attached report reference: _____

Signature: _____ Date: _____

Approval applies only to the passive-backplane scope and exact reviewed configuration. It does not approve absent daughtercard/power-module designs or commercial production.